On one hand, the product owner must understand the needs and priorities of the organizational stakeholders, the customers, and the users well enough to act as their voice. In this respect the product owner acts as a product manager, ensuring that the right solution is developed.

On the other hand, the product owner must communicate to the development team what to build and the order in which to build it. The product owner must also ensure that the criteria for accepting features are specified and the tests that verify those criteria are later run to determine whether the features are complete. The product owner doesn't write detail-level tests but ensures that the high-level ones are written so that the team can determine when the product owner will consider the feature complete. In these respects the product owner is part business analyst and part tester.

Principal Responsibilities

Figure 9.2 illustrates the principal product owner responsibilities.

This is clearly a full-time role with significant responsibilities. As a matter of fact, when you read the description that follows, you might start to think that it is not practical for one person to handle all of these responsibilities or to have all of the attributes necessary to be successful in the role. In most cases a single person can and should fill the product owner role; however, under certain circumstances, product owner teams or product owner proxies might be practical. Both concepts will be covered later in the chapter.

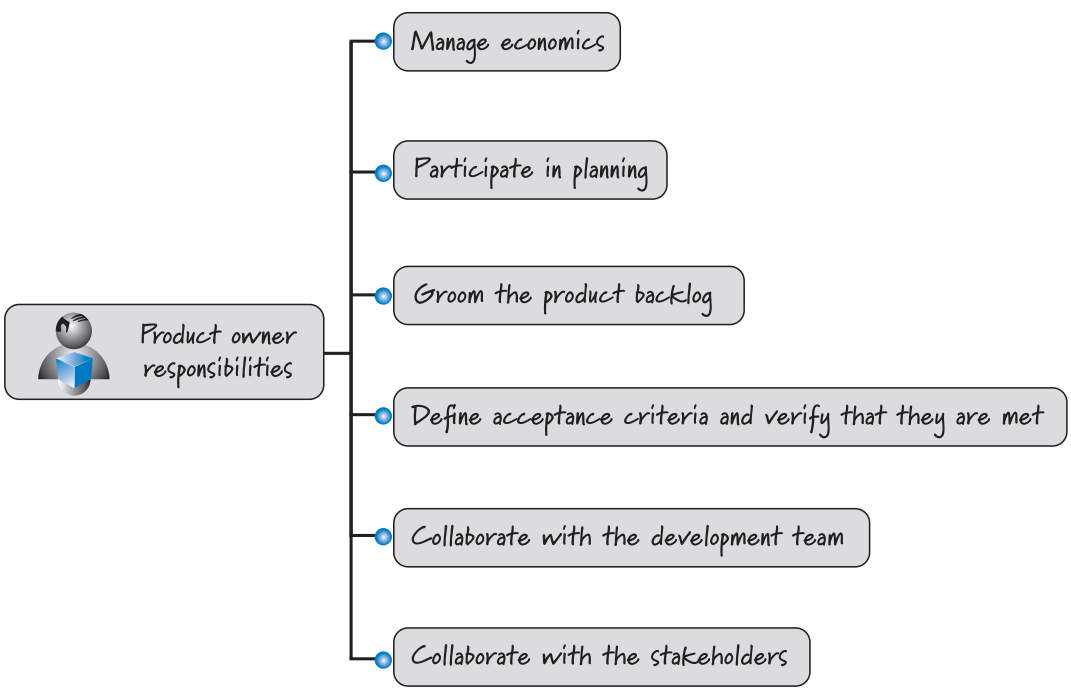


FIGURE 9.2 Principal product owner responsibilities

Manage Economics

The product owner is responsible for ensuring that good economic decisions are continuously being made at the release, sprint, and product backlog levels (see Figure 9.3).

Release-Level Economics

At the release level the product owner continuously makes trade-offs in scope, date, budget, and quality as a stream of economically important information arrives during product development. Trade-offs made at the beginning of a release might no longer be appropriate in the presence of new information that arrives during the release.

For example, what if several weeks into a six-month, fixed-date development effort we recognize an opportunity to increase revenue by 50% if we take one extra week (4% schedule slip) to add a newly identified feature to the release? Should we trade one week's time and the additional cost for the extra revenue? The product owner oversees this decision. In many cases he can unilaterally make the decision. Other times the product owner might recommend a decision but still work with others to secure their input (and at times approval) to execute the decision.

Also, at the end of every sprint the product owner oversees a decision as to whether or not to fund the next sprint. If good progress is being made toward the release goal or the next sprint is otherwise economically justified, the next sprint will be funded. If poor progress is being made or the economics don't support additional expenditures, the effort might be canceled.

A satisfied product owner might also oversee a decision to stop funding further development at the end of a sprint if the product is ready to ship and additional expenditures simply aren't justified. For example, let's say we planned for a ten-sprint release. After sprint 7, the product owner reviews the remaining product backlog items and concludes that the cost to create those items is greater than the value they deliver. The product owner might conclude that shipping the product early instead of

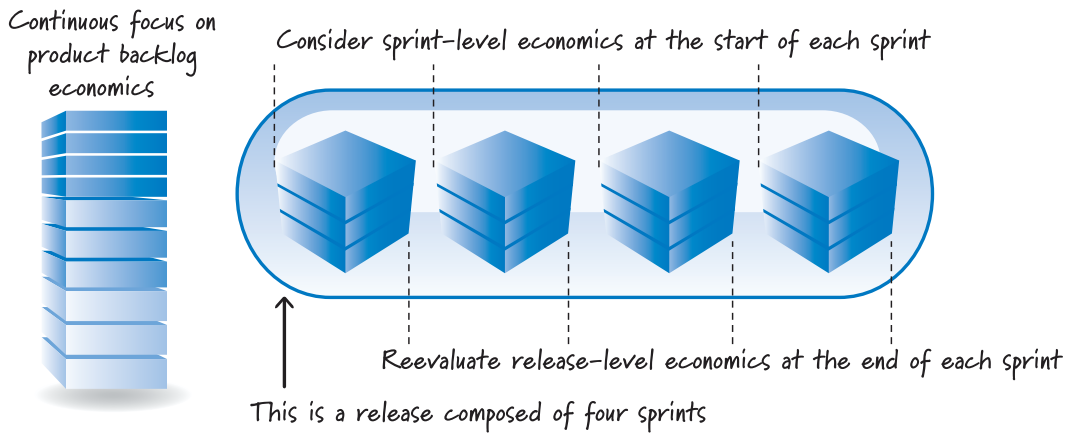


FIGURE 9.3 The product owner manages economics.

continuing with the original ten-sprint plan is economically more sensible. This flexibility to deliver early is enabled by ensuring that the higher-value items at the top of the product backlog are worked on first and the team is completing work each sprint in accordance with a strong definition of done.

And, of course, the product owner might also conclude that we should stop funding at the end of a sprint because core economic properties have changed. For example, what if we are creating a country-specific product and a regulatory agency in that country revises its laws, making it unprofitable or perhaps even illegal for us to sell the product? In cases like these, a product owner might oversee canceling the development effort even if things are otherwise going well.

Sprint-Level Economics

In addition to release-level economics, the product owner also manages sprint-level economics, ensuring that a good return on investment (ROI) is delivered from each sprint. Good product owners treat their organization's money as if it were their own money. In most cases the product owner knows the cost of the next sprint (the duration and team composition of the sprint are known). With this knowledge the product owner should ask himself at sprint planning, "Would I write a check out of my own bank account equal to the cost of this sprint to get the features that we're planning to build in this sprint?" If the answer is no, a good product owner wouldn't spend the organization's money either.

Product Backlog Economics

As I discussed in Chapter 6, the product owner is responsible for prioritizing the product backlog. When economic conditions change, the priorities in the product backlog will likely change as well.

For example, let's say that at the start of a release the product owner believes a feature is valuable to a large percentage of the target users and the team believes that only a modest effort is required to create it. After a few sprints, however, the team discovers that the feature will require a large effort to complete and is valuable to only a fraction of the target users. Because the cost/benefit ratio of this feature has dramatically changed, the product owner should reprioritize the product backlog to reflect this knowledge—perhaps by dropping the product backlog items associated with the feature.

Participate in Planning

The product owner is a key participant in the portfolio-, product-, release-, and sprint-planning activities. During portfolio planning (see Chapter 16), the product owner works with internal stakeholders (perhaps an approval committee or governance board) to position the product correctly in the portfolio backlog and to determine when to start and end product development. During product planning (see

Chapter 17), the product owner works with the stakeholders to envision the product. During release planning (see Chapter 18), the product owner works with the stakeholders and the team to define the content of the next release. During sprint planning (see Chapter 19), the product owner works with the development team to define a sprint goal. He also provides valuable input that enables the development team to select a set of product backlog items that the team can realistically deliver by the end of the sprint.

Groom the Product Backlog

The product owner oversees the grooming of the product backlog, which includes creating and refining, estimating, and prioritizing product backlog items (see Chapter 6). The product owner doesn't personally perform all of the grooming work. For example, he might not write all of the product backlog items; others might contribute them. The product owner also doesn't estimate the items (the development team does that) but is available for questions and clarification during estimation. The product owner is, however, ultimately responsible for making sure that the grooming activities take place in a way that promotes the smooth flow of delivered value.

Define Acceptance Criteria and Verify That They Are Met

The product owner is responsible for defining the acceptance criteria for each product backlog item. These are the conditions under which the product owner would be satisfied that the functional and nonfunctional requirements have been met. The product owner may also write acceptance tests corresponding to the acceptance criteria, or he could enlist the assistance of subject matter experts (SMEs) or development team members. In either case, the product owner should ensure that these acceptance criteria (and frequently specific acceptance tests) are created before an item is considered at a sprint-planning meeting. Without them, the team would have an incomplete understanding of the item and would not be ready to include it in a sprint. For this reason, many Scrum teams include the existence of clear acceptance criteria as an item on their definition-of-ready checklist (see Chapter 6).

The product owner is ultimately responsible for confirming that the acceptance criteria are met. Once again, the product owner may choose to run acceptance tests himself or may enlist the assistance of expert users to help confirm that the product backlog item meets the conditions of satisfaction. The team might help create a testing infrastructure that enables the product owner or the feature SMEs to run these tests more efficiently, but the product owner should be the final judge of whether an item meets expectations.

It is important for the product owner to verify acceptance criteria during sprint execution rather than waiting until the sprint review. By doing some testing as the features are completed, the product owner can identify mistakes and misunderstandings that the team can fix before the sprint review. Also, because the team is allowed

to demonstrate only completed features at the review, the product owner must ensure that acceptance tests are run prior to the review so that the team knows which features actually meet the definition of done.

Collaborate with the Development Team

The product owner must closely collaborate with the development team on a frequent basis. The product owner is an engaged, committed, everyday role. Many organizations just starting to adopt Scrum fail to foster adequate product owner engagement with the development team, delaying essential feedback and substantially reducing the value of that feedback when it does occur.

This failure to engage can also occur when people new to the product owner role assume that their level of involvement when using Scrum should resemble their involvement during phase-based development. Figure 9.4 compares the typical level of customer- or business-side engagement during a traditional sequential development effort with that expected of a product owner when using Scrum.

Using traditional, phase-based development, the pattern of engagement resembles a U-shaped or bathtub-shaped curve. Initially the customers have considerable upfront involvement to help define the complete set of requirements. Once the effort transitions into more technical phases (such as design, coding, and certain types of testing), the customers are "no longer needed." As such, their level of engagement is quite low or nonexistent during a majority of the effort. In fact, during traditional development, the customers don't reenter the process until near the very end, when they are required to perform user acceptance testing on what was built. What customers typically discover at this point is that what has been built isn't exactly what they

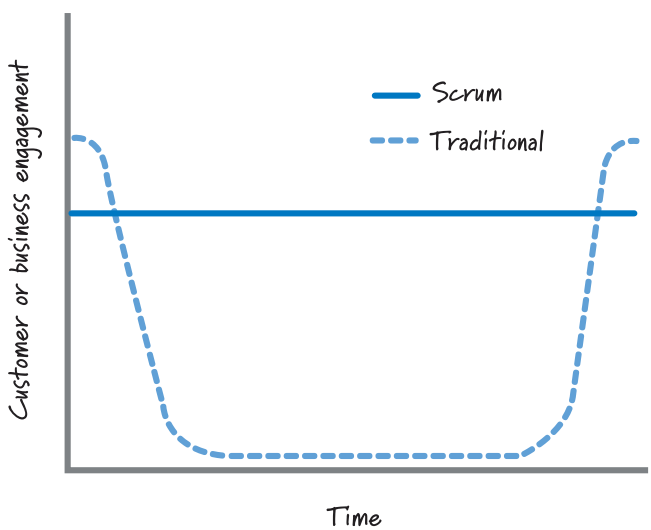


FIGURE 9.4 Comparison of customer or business engagement over time

wanted. To make matters worse, it's usually too late or too costly for them to make changes—at least in this release. Customers who come in expecting to be delighted instead leave surprised, frustrated, and disappointed. This is when the finger-pointing moves into high gear. The customer claims, "If you guys had read my requirements document more carefully, you would have built what I really wanted," and the development team retorts, "Well, if you had written your document more clearly, we would have built something different. We built what you asked for!"

Using Scrum, we build a feature at a time, not a phase at a time. This means that we perform all activities to create a particular feature (design, code, integrate, test) during one sprint. Therefore, a constant high level of engagement by the product owner is essential. With such close interaction on short, timeboxed iterations there is far less chance for the product owner and the development team to become disconnected. A secondary benefit is that there is no finger-pointing when doing Scrum well!

Collaborate with the Stakeholders

The product owner is the single voice of the entire stakeholder community, internal and external. **Internal stakeholders** can include business systems owners, executive management, program management, marketing, and sales. **External stakeholders** can include customers, users, partners, regulatory bodies, and others. The product owner must work closely with the entire stakeholder community to gather input and synthesize a coherent vision to guide product development.

If the product owner becomes overwhelmed and spreads himself too thin, it will be difficult for him to collaborate with both the development team and the stakeholders at the required level. In some circumstances the workload may be more than one person can reasonably perform, in which case the product owner may enlist the assistance of others to help fulfill the responsibilities of the role. I will address this later when I discuss the concept of a product owner team.

Characteristics/Skills

Figure 9.5 illustrates important characteristics of the product owner role.

Although there are numerous characteristics that I look for in a good product owner, they can be grouped into four categories: domain skills, people skills, decision making, and accountability.

Domain Skills

The product owner is a visionary who can synthesize a product vision and lead the team to achieve that vision. Having a vision doesn't mean that every detail of the vision or the path to achieving the vision is perfectly clear. A good product owner knows that not everything can be anticipated up front and is willing to adapt when change is warranted.

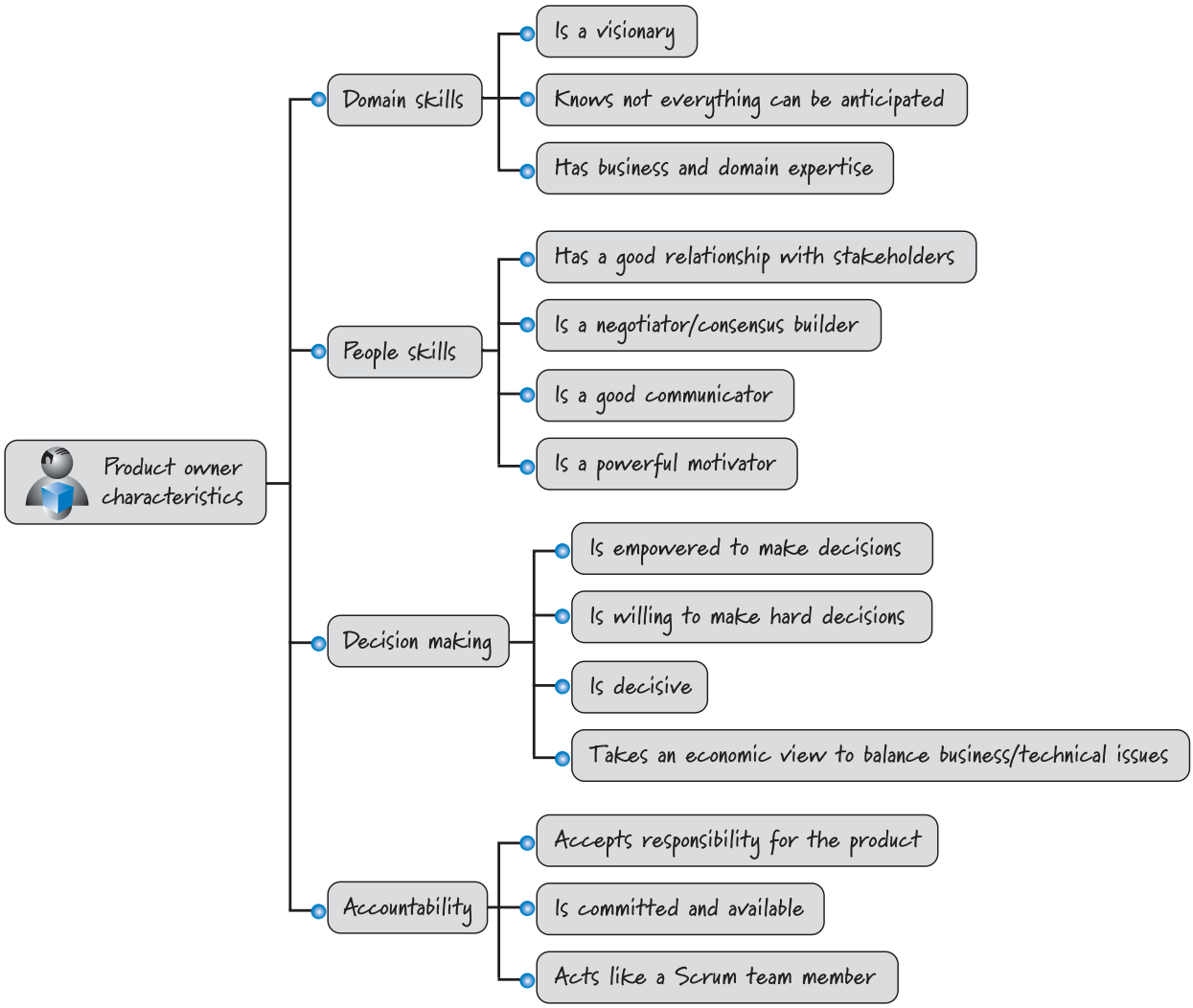


FIGURE 9.5 Product owner characteristics

To be effective at vision creation and execution, a product owner must have appropriate business and domain knowledge. It's difficult to be an effective product owner if you're new to the product domain. How can you set priorities among competing features if you don't know the subject matter?

People Skills

A product owner must also be the "voice of the customer," which requires a good relationship with stakeholders. Because there are frequently multiple stakeholders who could have conflicting needs, the product owner must also be a good negotiator and consensus builder.

The product owner is the linchpin between the stakeholder community and the rest of the Scrum team. In this position, the product owner needs good

communication skills to work with both constituencies and to convey information to each group in the appropriate language. A good communicator also exhibits the following qualities: is willing to speak up even if doing so goes against the status quo; confident in his ideas; knowledgeable of the subject matter; able to communicate in a simple, concise, and easily understood way; and credible.

A product owner is also a powerful motivator. When the going gets tough, the product owner can remind people of why they are investing the effort and help people maintain an enthusiastic outlook by reinforcing the business proposition.

Decision Making

The product owner must be empowered to make decisions. A frequent impediment to organizations new to Scrum is that the person selected to be the product owner is not empowered to make any significant decisions. Such a person is not the product owner.

The product owner must also be willing to make the hard decisions—usually by trading off constraints like scope, date, and budget. These decisions must be made in a timely manner and should not be reversed without good reason. In other words, the product owner should be a decisive decision maker.

In making these decisions, a product owner must maintain the proper balance between business needs and technical realities. Although the Scrum team as a whole is responsible when systems accrue unacceptable levels of technical debt, naive product owner decisions and ones that fail to consider system-level effects can frequently be a significant contributing factor.

Accountability

The product owner is held accountable for delivering good business results. This accountability doesn't absolve other Scrum team members from their own accountability to participate in generating a good return on investment. However, the product owner is on the hook for making sure the resources are being used in an economically sensible way and must accept responsibility if they are not. After all, the product owner has many opportunities along the way to change the product backlog, readjust priorities, or even oversee canceling the development effort entirely.

The product owner must be committed and available to both the stakeholders and the rest of the Scrum team. Being a product owner is a full-time job; to try to do it part-time is a recipe for failure.

Finally, the product owner is a member of the Scrum team and therefore realizes that good economic results are impossible without the collaborative efforts of the full Scrum team. So, the product owner treats the development team and ScrumMaster with respect and trusts that they are partners in delivering the desired results. Together, all Scrum team members should share a Musketeer attitude (I will describe this concept further in Chapter 11). There is no us-versus-them attitude. The product owner, ScrumMaster, and development team are one unit working together toward the same goal.

A Day in the Life

To better appreciate the scope of product owner responsibilities, let's look at a "day in the life" of a product owner across part of a product development effort (see Figure 9.6).

During weeks 1 and 2 the product owner is engaged in both portfolio planning (see Chapter 16) and product planning (see Chapter 17). As part of portfolio planning, the product owner might work with the portfolio manager or the governance

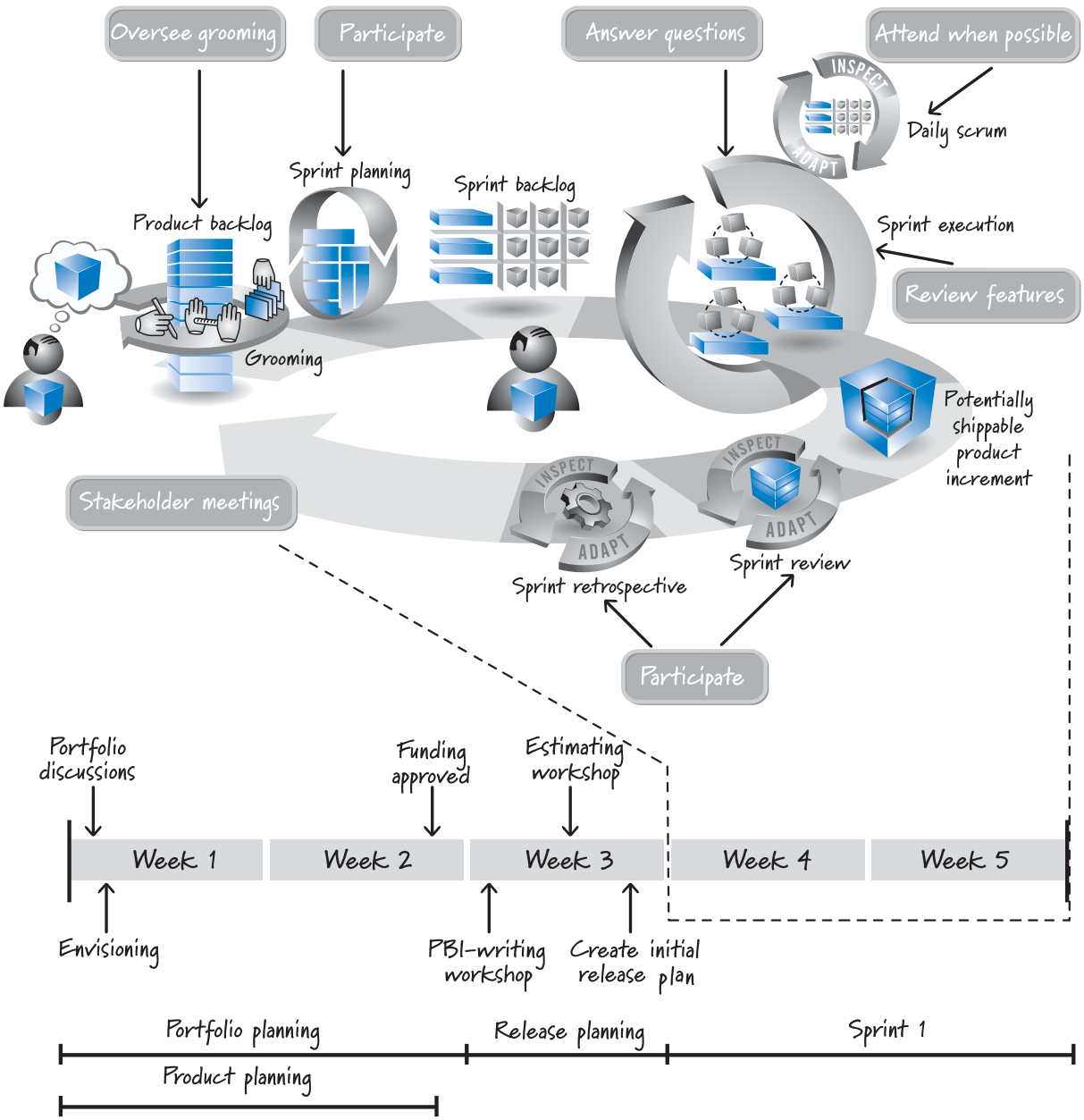


FIGURE 9.6 A day in the life of a product owner

board to discuss portfolio expectations that could influence the new-product planning. Those discussions provide input to **product planning**, where the product owner, working with appropriate stakeholders and others, will perform the envisioning of the new product.

At the completion of product planning, the proposed product will be submitted to portfolio planning, where it is subjected to the organization's **economic filter** to determine if development will be funded and when work can begin. Figure 9.6 shows all of this occurring immediately after product planning is complete; in many organizations there would likely be a delay between the end of envisioning and when the approval committee or governance board would review and approve funding and work would begin.

In week 3 the product owner is engaged in initial release planning (see Chapter 18). This typically involves a PBI-writing (story-writing) workshop (see Chapter 5 for more details) that includes internal stakeholders, the development team members, and possibly external stakeholders to generate a high-level product backlog that can be used during release planning. The development team members should be available to participate because funding has already been approved. If necessary, a surrogate team can be used if the development team is not yet formed.

Following the PBI-writing workshop, the product owner participates in an estimating workshop (probably a series of meetings over a day or two) during which development team members (or a surrogate team if the actual team has not yet been assigned) estimate the size of high-value product backlog items.

Next, the product owner facilitates an initial release-planning session (longer-term planning). Because some number of product backlog items have already been estimated, the focus of this release-planning activity is on prioritizing the product backlog and balancing the constraints of scope, schedule, and budget (see Chapter 18). The stakeholders are the principal co-participants in this activity; however, some or all of the development team members will need to be involved at some point to identify technical dependencies that could affect the order of the product backlog items.

The goal is to do a sufficient amount of release planning to have an acceptable level of clarity of the overall release, and to provide initial answers to business questions such as what will be delivered and when. For most products this activity should take no more than a day or two to complete. As I will discuss in Chapter 18, release planning is an ongoing activity, so we shouldn't overinvest at this point by trying to be very precise; we'll update the release plan as better information becomes available.

Following release planning, the Scrum team performs the first sprint (Figure 9.6 shows a two-week sprint during weeks 4 and 5). At the start of the sprint the product owner oversees the sprint-planning activity (see Chapter 19). During sprint execution (see Chapter 20), the product owner tries to attend the team's daily scrums; this may not always be possible, but it is a good practice. During the daily scrum, the product owner listens to better understand how the current sprint is progressing and to identify opportunities to assist the development team. Perhaps a team member mentions

he is a little fuzzy on the specifics of a product backlog item and needs some clarification before he can complete his current task. If it is a quick clarification, the product owner might offer it up during the daily scrum. If the answer requires anything other than a few-second response, the product owner should say, "I'd be happy to stick around after the daily scrum and discuss it with you."

The product owner must also be available (typically every day) to answer questions and to test features as they become reviewable. If the product owner knows he can't be available every day to perform these responsibilities, he must delegate them to an appropriate person so that the development team will not be blocked. I will discuss this idea further later in this chapter.

Also during sprint execution the product owner meets with both the internal and external stakeholders to ensure that priorities for the upcoming sprint are correct and to secure valuable user input that will affect the features chosen for future sprints.

The product owner also performs frequent product backlog grooming, which includes writing new product backlog items and refining existing items, then working with the team to get them estimated, and the stakeholders and team to get them prioritized.

At the end of the sprint, the product owner participates in the two end-of-sprint inspect-and-adapt activities: the sprint review (see Chapter 21) and the sprint retrospective (see Chapter 22). After these are completed, the sprint cycle repeats and the product owner participates in the next sprint-planning activity.

Who Should Be a Product Owner?

Most non-Scrum organizations probably won't have an existing role labeled "product owner." So, who within the organization should fill the product owner role?

As I mentioned earlier in this chapter, the product owner needs to face two directions simultaneously: toward the internal and external stakeholders and toward the development team. Thus, the product owner role is a melding of authority and responsibilities that have historically been found in several traditional roles. In its most encompassing expression, a product owner incorporates elements of the roles of product manager, product marketer, project manager (discussed further in Chapter 13), business analyst, and acceptance tester.

Exactly who should be the product owner depends on the type of development effort and the specific organization. Table 9.1 suggests good candidates for the product owner role for different types of development.

Internal Development

On an internal development effort, an empowered person from the group that will benefit from the development should be the product owner. For example, if an internal IT group develops a system for the marketing group, an empowered person from the marketing team should be the product owner (see Figure 9.7).

TABLE 9.1 Product Owners for Different Types of Product Development

Development Type	Candidate Product Owner
Internal development	Representative/customer from the business area benefiting from the solution
Commercial development	Internal proxy for the actual customers and users (typically a product manager, product marketer, or project manager)
Outsourced development	Representative/customer from the company paying for the solution and receiving the benefits
Component team (architectural development)	Typically a technical person who can best prioritize the backlog of technical items

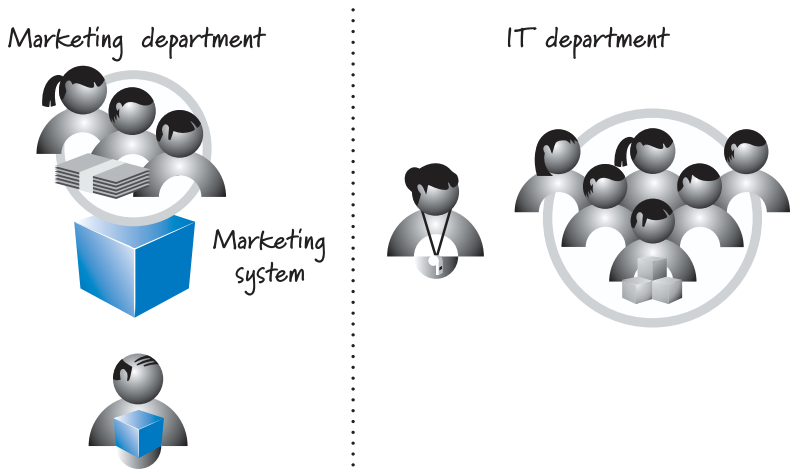


FIGURE 9.7 Example of a product owner on internal development

Some organizations (typically those that haven't yet learned the importance of having a business person as the day-to-day engaged product owner) might ask an IT person to handle the day-to-day product owner responsibilities. I will review the issues with this approach when I discuss the concept of a product owner team later in this chapter.

Commercial Development

On a commercial development effort—for example, a company building a product for sale to external customers—the product owner should be an organization employee who acts as the voice of the actual customers. Frequently this person comes from the ranks of product management or product marketing (see Figure 9.8).

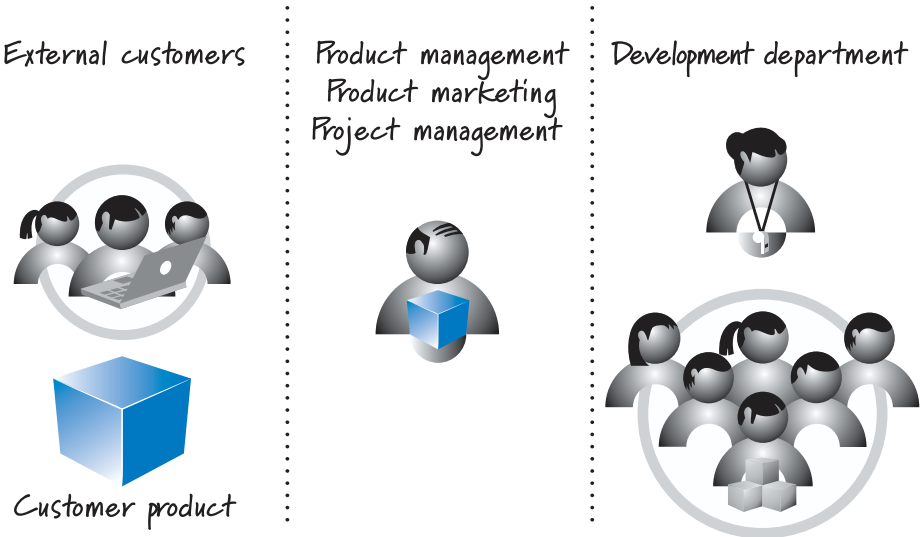


FIGURE 9.8 Example of a product owner on commercial development

Scrum practitioners have hotly debated whether or not the product owner role is really just the Scrum (and agile) renaming of the product *manager* role. Some believe the two roles are synonymous. Others advocate that the product owner role is larger than the product manager role. And, of course, still others argue that the product manager role is larger. Here's how I view it.

The areas of product management and product marketing are quite expansive. Pragmatic Marketing, Inc., a well-known and respected company in the product management/marketing field, has created a highly regarded framework that defines the roles and responsibilities for technology product management and product marketing teams (see Figure 9.9).

To cover all of these activities, Pragmatic Marketing suggests that multiple roles are needed, including product strategy champions, technical product managers, and marketing product managers. Most people would agree that if a commercial organization needs to perform all of these activities for a larger product, a team of people is likely necessary.

Is the product owner expected to perform all of these activities? Those who believe the product owner role is a subset of the traditional product manager role argue that a product owner is really just the "technical product manager" and therefore would focus primarily on the small number of activities shown inside the dashed line in Figure 9.9. They believe that because the product owner has to be available to the team on a day-to-day basis, he would not have time to focus on the other activities.

Certainly the product owner is responsible for performing the activities within the dashed line, but I believe that the product owner role is also responsible for more

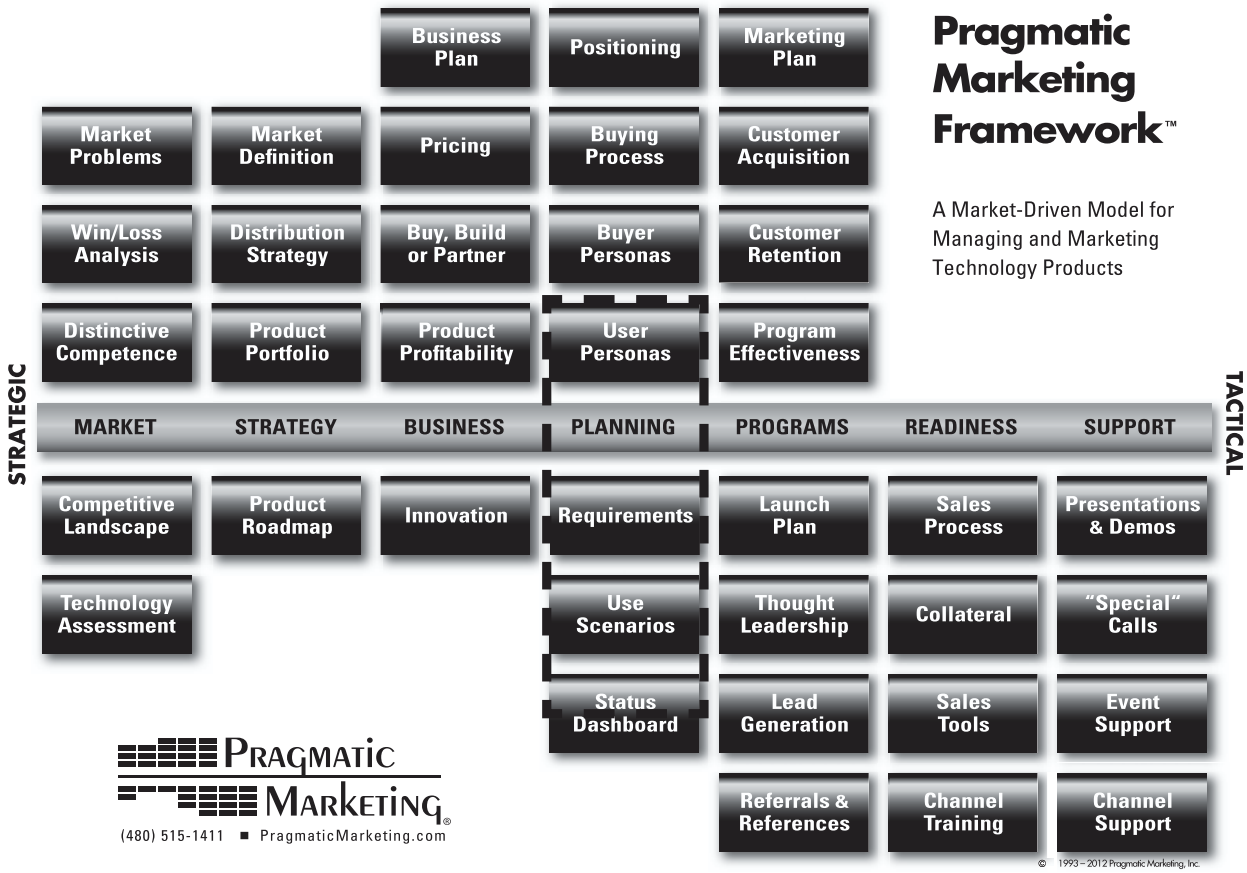


FIGURE 9.9 Pragmatic Marketing framework

activities. In fact, I believe that the product owner *role* should be responsible for performing as many of the activities shown in Figure 9.9 as are necessary and practical for the product owner to perform. The extent of that responsibility would depend on the organization, the specific product, and the skills of the person selected to be the product owner. For example, an organization that is producing a simple unit-conversion application for sale in a mobile device app store will not require as many activities as an organization creating the next major release of its enterprise Business Intelligence product. Therefore, it isn't practical to universally define the extent of the product owner's responsibility relative to the Pragmatic Marketing framework.

As I will discuss shortly, there are times when the scope of the product owner activities may be too large for any one person to adequately perform. In such cases, we might have a product owner team that includes people who focus on strategy and marketing. However, there will always be a single individual who functions in the product owner role for a Scrum team.

Outsourced Development Project

On an outsourced development effort—for example, company A contracting with company B to build a solution—a representative from company A should be the product owner. Company B may assign an internal person to closely liaise with the product owner, but the product owner should be from the company that is paying for the solution and receiving the benefits (see Figure 9.10).

The product owner role gets complicated if company A and company B enter into a traditional fixed-price development contract. In that case company B will almost certainly feel that it should be fulfilling most of the product owner responsibilities because it has assumed the risk of a fixed-price contract. In reality, though, company A, as the actual customer, should fill the product owner role. A more appropriate contract would have company A leasing the high-performance development team and ScrumMaster from company B and company A providing the product owner.

Component Development

Last, some organizations might use component teams (see Chapter 12) that build parts of customer solutions but not full customer solutions. These teams tend to create components or other assets that are then reused by other teams to assemble customer-valuable solutions. Because these teams focus at a technical component level, their product owners are typically technically oriented people who are capable of defining and prioritizing the technical features in their backlogs (see Figure 9.11).

In the figure, there are three business-oriented feature teams that create features that are valuable to the end users. Each feature team has its own product owner

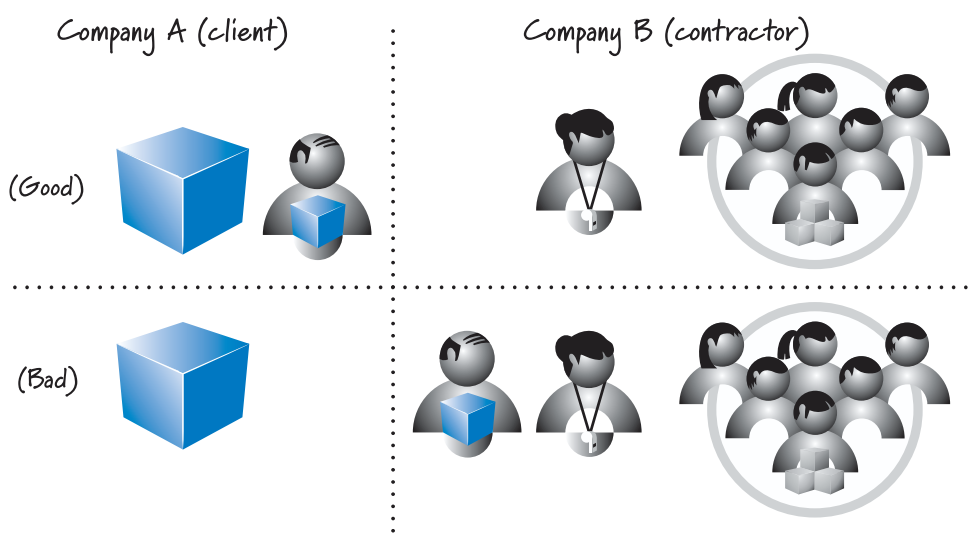


FIGURE 9.10 Example of a product owner on outsourced development

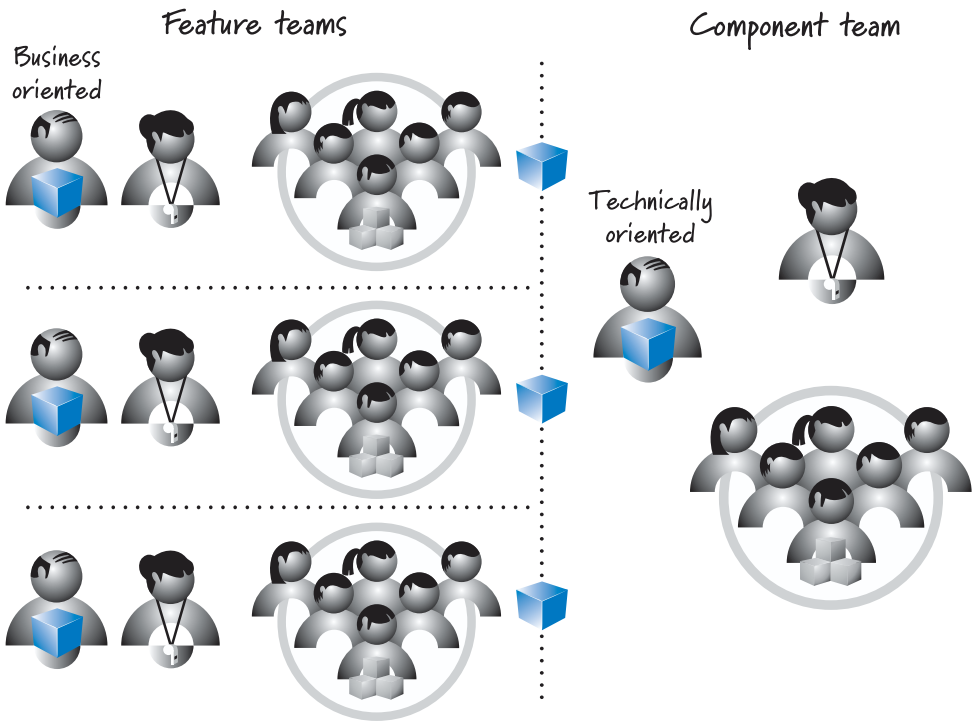


FIGURE 9.11 Example of a product owner on component development

focused on that team's features. Each of these feature teams also leverages the work of a component team that provides them with an asset necessary to complete their features. The component team needs a product owner who can prioritize and oversee the development of the various component-level requests being made by the feature teams. The component team's product owner is likely to be more technically oriented than the product owners of the feature teams.

Product Owner Combined with Other Roles

If capacity permits, the same person may act as the product owner for more than one Scrum team (see Figure 9.12).

Usually it is easier for this person to be the product owner of multiple teams on the same development effort, because the work of these teams will most likely be highly interrelated.

Although there are times when the same individual may be the product owner and a member of the development team, it is considered a bad idea for the same person to be both the product owner and the ScrumMaster on the same Scrum team. These two roles counterbalance each other; having one person play both roles creates a conflict of interest that we should try to avoid.

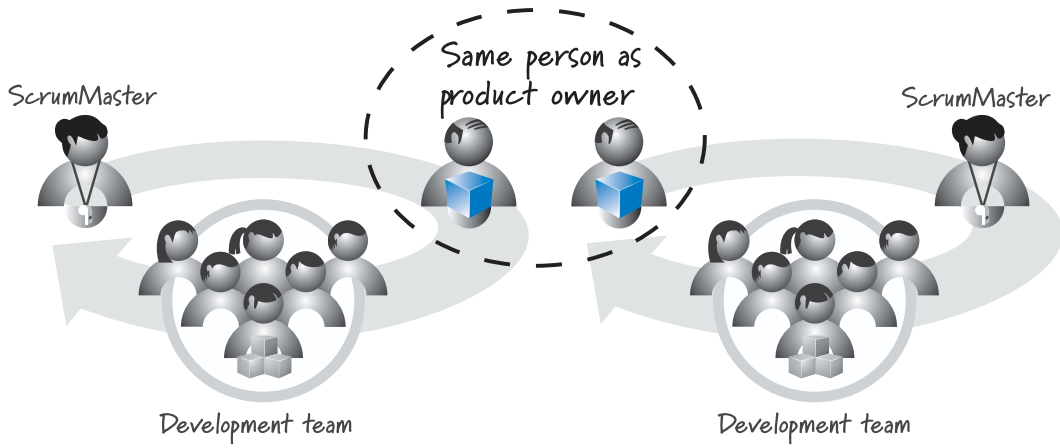


FIGURE 9.12 Same person as product owner of more than one Scrum team

Product Owner Team

Every Scrum team must have a single person who is identified as the product owner and is the only person empowered and accountable for fulfilling the product owner responsibilities for that Scrum team.

Should we allow a team of people to perform the product owner role? If by team we mean a group of people with shared decision making and accountability, definitely not. To properly apply Scrum, we need one individual to be *the* product owner, making decisions and acting as the single voice of the stakeholder communities to the Scrum team.

That being said, some organizations might form what they call a "product owner team" because they recognize that, in their circumstances, the product owner cannot do the job without a select group of people to provide input and guidance. In other companies, the workload of being a product owner might be greater than any one full-time person can reasonably perform. In those cases, the product owner delegates some product owner responsibilities to other people. Forming a product owner team in either of these circumstances is acceptable as long as there is one person on the team who is the ultimate decision maker and as long as having a product owner team does not degrade into design by committee, with every decision needing approval from eight other people.

Be careful when creating product owner teams. Product owners who are not properly skilled to be the empowered central point of product leadership don't need a committee—they need a different role. Similarly, product owners who are too busy to fulfill their responsibilities might not need a team. Perhaps the real problem is that the organization has chosen to start too many development efforts at one time, or there are just too few product owners to cover the necessary products.

Alternatively, perhaps the product we are building is just too large and should be broken into a series of smaller pieces with more frequent releases. With small pieces a single person might more easily fill the product owner role. Also, if we have structured our teams poorly (see Chapter 12), or poorly conceived the product backlog structures (see Chapter 6), a single product owner might find it difficult to do his job. Be sure that your product owner teams are truly necessary and not just masking an underlying problem; otherwise, the situation will become more complicated and jeopardize your overall outcome.

Product Owner Proxy

As I mentioned earlier, some companies doing internal development ask an IT person (for example, a business analyst or development manager) to be the product owner because the business unit person is too busy with other work. Because everyone knows that the IT person is really not empowered to make important final decisions (one of the key responsibilities of any product owner), organizations that do this have filled the product owner role ineffectively and confusingly. A better solution is to free up enough of the business unit person's time to be the true product owner but have the IT person act as the product owner proxy in certain interactions with the team.

A **product owner proxy** is a person enlisted by the product owner to act on his behalf in particular situations. Everyone on the Scrum team knows that the proxy is not the actual product owner, but everyone also knows that the product owner has empowered the proxy to make at least some tactical decisions on his behalf. A common example is when the product owner spends a great deal of time meeting with customers and users to make sure he has his fingers on the pulse of the marketplace. This person is reliably unavailable to the development team on a day-to-day basis. In this case, the product owner might enlist the support of a proxy to handle the day-to-day interaction with the development team regarding product backlog items.

For this approach to work, it is essential that the product owner actually empower the proxy to make decisions and not unreasonably overrule those decisions in a way that would undermine the proxy's credibility with the team. Remember, even though the product owner can empower others to assist him, he cannot delegate the ultimate responsibility for ensuring that the work gets done—he is still on the hook for that.

Chief Product Owner

Another situation where a product owner team is frequently created is on very large products. Earlier I noted that a single person might be the product owner for a couple of Scrum teams, but what about scenarios that involve many teams? For example, I trained and coached an organization that had a development effort that involved upward of 2,500 people. With an average team size of fewer than 10 people, the organization had more than 250 teams on the effort. One person can't be the product owner for 250 teams. In fact, one person can't be the engaged, day-to-day product

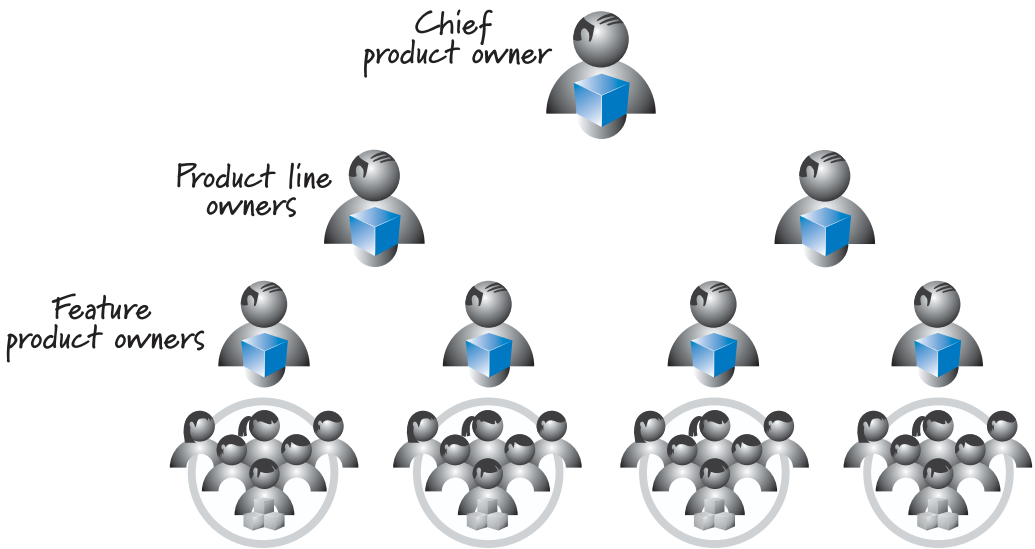


FIGURE 9.13 Hierarchical product owner role

owner for more than a few teams. In cases like these, the product owner role needs to scale hierarchically as shown in Figure 9.13.

Ultimately in Figure 9.13 the person labeled **chief product owner** is *the* product owner for the whole product. However, the chief product owner has a team of product owners to ensure that the product owner role is filled correctly at each lower level in the hierarchy. If you choose to use this approach, ensure that the individual team product owners remain empowered to make the vast majority of the decisions at their levels, rather than having to pass such decisions up the hierarchy to be made at higher levels.

Closing

In this chapter I expanded the description of the product owner role. I emphasized this role as the empowered central point of product leadership and described important responsibilities and characteristics of the role. I then described what the product owner does during the various Scrum activities on a project. I went on to discuss what type of person should fill the role for different project types. Then I described how a single person might be the product owner for more than one Scrum team, and how on occasion a single person might be both a product owner and a team member on the same Scrum team. I ended by discussing the idea of a product owner team with a focus on product owner proxies and chief product owners. In the next chapter I will discuss the ScrumMaster role.

Chapter 10
ScrumMaster

In this chapter I describe the ScrumMaster role. I begin by describing the purpose of the role relative to the other Scrum roles. Then I define the principal responsibilities and characteristics of a ScrumMaster. Next I illustrate a "day in the life" of the ScrumMaster, which leads to a discussion of whether or not the ScrumMaster role is full-time. I end by describing the kind of person who typically fulfills the ScrumMaster role.

Overview

The ScrumMaster is one of the three roles that constitute every Scrum team (the others being the product owner and the development team). While the product owner is focused on building the right product and the development team is focused on building the product right, the ScrumMaster is focused on helping everyone understand and embrace the Scrum values, principles, and practices. The ScrumMaster acts as a coach to both the development team and the product owner. A ScrumMaster also provides process leadership, helping the Scrum team and the rest of the organization develop their own high-performance, organization-specific Scrum approach.

Principal Responsibilities

Figure 10.1 illustrates the principal ScrumMaster responsibilities.

Coach

The ScrumMaster is the agile coach for the Scrum team—both the development team and the product owner (see Adkins 2010 for a comprehensive description of an agile coach). By coaching both roles, the ScrumMaster can remove barriers between the roles and enable the product owner to directly drive development.

Analogous to the coach of a sports team, the ScrumMaster observes how the team is using Scrum and does anything possible to help it get to the next level of performance. When problems arise that the team can and should be able to solve, the ScrumMaster's attitude, like that of any good coach, is "I'm not here to solve your problems for you; instead, I'm here to help you solve your own problems." If the problem is an impediment that the team can't resolve, the ScrumMaster takes ownership of getting it resolved.

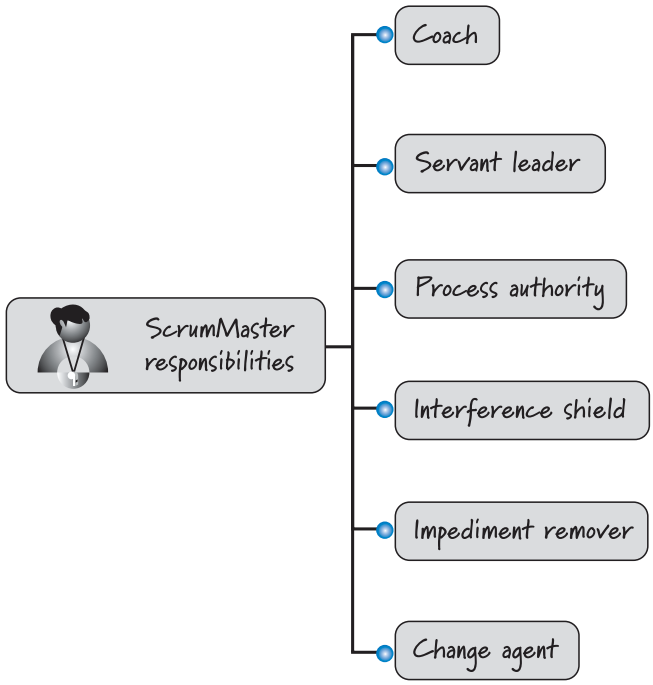


FIGURE 10.1 Principal ScrumMaster responsibilities

The ScrumMaster coaches a new product owner by helping him understand and perform his product owner responsibilities. Once the ScrumMaster helps the product owner get established in his role, she provides him with ongoing assistance for activities such as grooming the product backlog. Furthermore, in keeping with the sports team analogy, the ScrumMaster's relationship with the product owner is very much like a sports team coach's main role with the team's owner: help the owner maximize business outcomes using Scrum, manage expectations, make sure the owner is providing the team with what it needs, and listen to the owner's complaints and requests for change and translate those into actionable improvements for the team.

Servant Leader

The ScrumMaster is often described as a **servant leader** of the Scrum team. Even when acting as the team's coach, the ScrumMaster is first and foremost a servant to the Scrum team, ensuring that its highest-priority needs are being met. A servant leader would never ask, "So, what are you going to do for me today?" Instead, a servant leader asks, "So, what can I do today to help you and the team be more effective?"

Process Authority

The ScrumMaster is the Scrum team's process authority. In this capacity, the ScrumMaster is empowered to ensure that the Scrum team enacts and adheres to the Scrum

values, principles, and practices along with the Scrum team's specific approaches. The ScrumMaster continuously helps the Scrum team improve the process, whenever possible, to maximize delivered business value.

Authority in this context is not the same type of authority that a functional manager or project manager would have. For example, the ScrumMaster doesn't hire and fire and cannot dictate to the team what tasks it should do or how to do them. The ScrumMaster also is not responsible for making sure the work gets done. Instead, the ScrumMaster helps the team define and adhere to its own process for making sure the work gets done.

Interference Shield

The ScrumMaster protects the development team from outside interference so that it can remain focused on delivering business value every sprint. Interference can come from any number of sources, from managers who want to redirect team members in the middle of a sprint, to issues originating from other teams. No matter what the source of the interference, the ScrumMaster acts as an interceptor (fielding inquiries, addressing management, and arbitrating disputes) so that the team can focus on delivering value.

Impediment Remover

The ScrumMaster also takes responsibility for removing impediments that inhibit the team's productivity (when the team members themselves cannot reasonably remove them). For example, I observed a Scrum team that was consistently unable to meet its sprint goals. The impediment was unstable production servers that the team used during testing (as part of its definition of done). The team itself had no control over these servers—that was the responsibility of the VP of Operations. Because the team itself could not remove the impediment, the ScrumMaster took ownership of improving the server stability by working with the VP of Operations and others who could actually do something about the stability issue.

Change Agent

The ScrumMaster must help change more than faulty servers and similar impediments. A good ScrumMaster must help change minds as well. Scrum can be very disruptive to the status quo; the change that is required to be successful with Scrum can be difficult. The ScrumMaster helps others understand the need for change, the impacts of Scrum outside of the Scrum team, and the broad-reaching benefits Scrum can help achieve. The ScrumMaster also ensures that effective change is occurring at all levels of the organization, enabling not only short-term success but, more importantly, the long-term benefits from using Scrum. In large organizations, the ScrumMasters might band together to become a more effective force for change.

Characteristics/Skills

Figure 10.2 illustrates important ScrumMaster characteristics.

Knowledgeable

To be an effective process coach, the ScrumMaster must be very knowledgeable about Scrum. The ScrumMaster should also understand the technical issues the team needs to address and technologies the team will use to create solutions. A ScrumMaster doesn't need to have tech-lead- or dev-lead-level knowledge, but reasonable technical knowledge is an asset. The ScrumMaster also doesn't need to be an expert in the business domain (the product owner does), but again, working knowledge of the business domain is very helpful.

Questioning

ScrumMasters use their coaching skills in conjunction with their process, technical, and business knowledge to ask great questions. They engage in intentional inquiry, asking the kinds of questions that make people stop and say, "Hmm. I never thought about that. Now that you ask that question, it makes me think there might be another way to go." Great ScrumMasters almost never directly answer a question but instead reflexively answer with their own question—not an annoying question, or a question

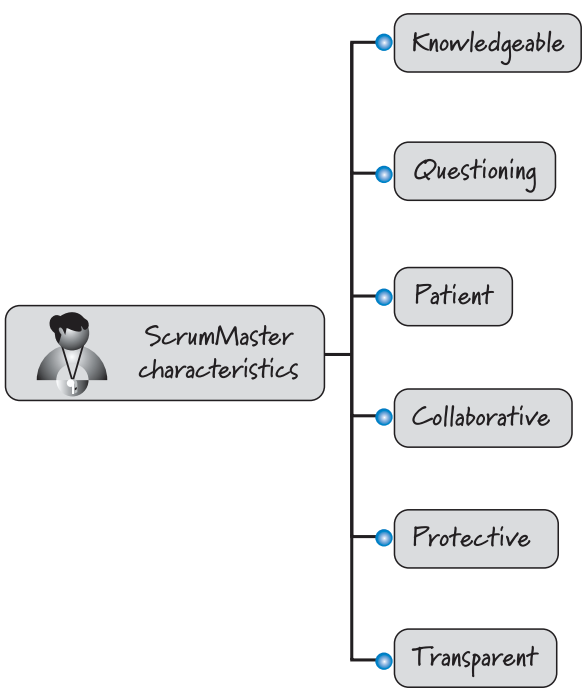


FIGURE 10.2 ScrumMaster characteristics

for the sake of asking a question, but rather a thoughtful, deep, probing question—thereby helping people realize that they have the insight to find their own answers (a form of Socratic questioning).

Patient

Because ScrumMasters prefer not to give out answers, they need to be patient, giving teams time to arrive at appropriate answers on their own. At times it is hard for me to be a ScrumMaster because I see the issue the team is dealing with and I "know" the answer. Well, at least I *think* I know the answer! It is arrogant for me (or any ScrumMaster) to believe that I am smarter than the collective intelligence of the team. So, at times I just have to bite my tongue and be patient, letting the team work out the solution, periodically asking probing questions to help guide things along.

Collaborative

The ScrumMaster must have excellent collaboration skills to work with the product owner, development team, and all the other parties, even those who might not be directly involved with Scrum. Also, as the process coach, the ScrumMaster is always looking for opportunities to help the Scrum team members achieve an enviable level of intra-team collaboration. A ScrumMaster can assist in this effort by personally exhibiting effective collaboration skills.

Protective

The ScrumMaster should be very protective of the team. The common analogy is that the ScrumMaster acts like a sheepdog, guarding the flock from wolves that might try to attack. In our context wolves could be organizational impediments or people with differing agendas. The ScrumMaster is adept at ensuring the protection of the team within the greater context of making economically sound business decisions. With acute sensitivity toward both team protection and business needs, the ScrumMaster helps the Scrum team achieve a healthy balance.

The ScrumMaster also helps team members who begin to wander away from the flock. When things get difficult, it is easy for people to fall back on familiar, non-agile approaches. In this case it is the ScrumMaster's job to help shepherd straying team members, helping them overcome their difficulties by reinforcing how to use Scrum more effectively.

Transparent

Finally, the ScrumMaster is transparent in all forms of communication. When working with team members, there is no room for hidden agendas; what you see and hear from the ScrumMaster must be what you get. People expect nothing less of a servant leader. The ScrumMaster also promotes transparent communication outside of the

Scrum team. Without transparent access to information it is difficult for the organization to inspect and adapt to achieve its desired business results from using Scrum.

A Day in the Life

What exactly is life like for a ScrumMaster during a sprint? Figure 10.3 is indicative (not a precise statement) of how much time the ScrumMaster of a newly formed team might spend doing each of the activities throughout a sprint. The percentage allocations would be different for a ScrumMaster of a high-performing Scrum team that has been working together for several years.

As illustrated in the figure, the ScrumMaster spends time each day organizing and facilitating the Scrum activities, including sprint planning, sprint execution, sprint reviews, sprint retrospectives, and daily scrums. This includes setting up the activities, overseeing their execution, and enabling the rest of the Scrum team to perform at a level where high-value results are achieved.

The ScrumMaster also spends time each day coaching the team members to help them improve their use of Scrum and technical practices. The ScrumMaster might also perform refresher training—for example, reminding a new team about the rules of Planning Poker when estimating product backlog items. Also, some amount of each day is dedicated to communicating (for example, updating sprint and release burndown or burnup charts, or discussions with non-Scrum-team members).

Throughout the sprint, the ScrumMaster spends some time working with the product owner on product backlog grooming activities (for example, writing and prioritizing new product backlog items). The ScrumMaster also works with the product owner to ensure that economically viable trade-offs are being made regarding important variables such as feature, date, budget, and quality.

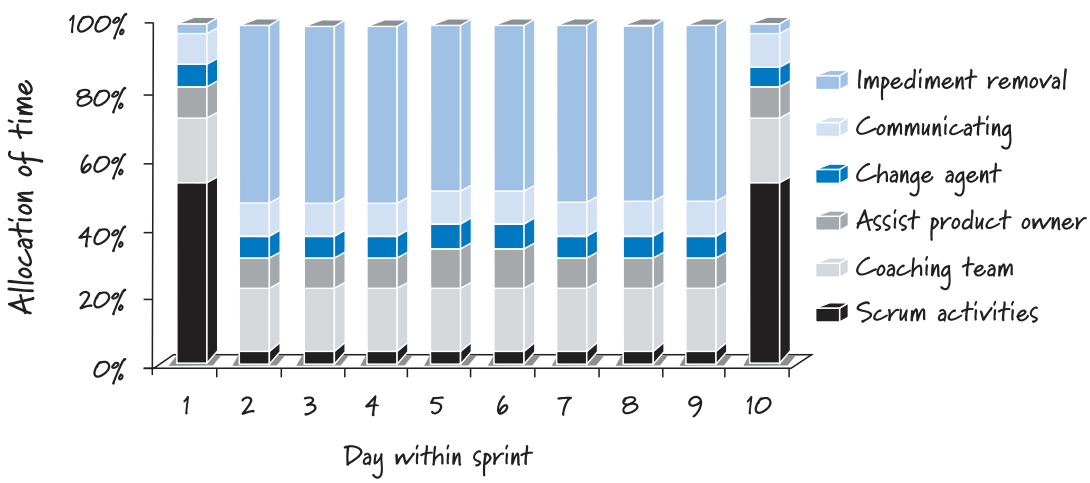


FIGURE 10.3 A day in the life of a ScrumMaster

The ScrumMaster also spends time acting as a change agent to help the organization better embrace Scrum throughout the value chain (from sales, marketing, HR, subcontracting, and so on).

The ScrumMaster spends a variable amount of time each day removing impediments. She might budget a fixed amount of time each day specifically for impediment removal. Of course, impediments can appear at any time, and they might be large and time-sensitive, so the ScrumMaster might need to dynamically reallocate time from other activities to address them.

Most teams and organizations that are new to Scrum will have many impediments when starting out and tend to focus on the ones that are obvious and somewhat easy to remove. That doesn't mean, however, that all impediments will be easily dispatched. In fact, the next level of impediments is often much more difficult and time-consuming to address. Impediment removal is a big variable in the ScrumMaster's day; it could easily change the allocations of time shown in Figure 10.3.

Fulfilling the Role

When considering the ScrumMaster role, we need to decide who is best suited for it, whether the role is a full-time job, and whether it can be combined with other Scrum and non-Scrum roles. Let's consider each of these.

Who Should Be a ScrumMaster?

Organizations new to Scrum won't have people in a role called ScrumMaster. So where do we find the ScrumMasters? I've seen great ScrumMasters come from many different existing roles. Some ScrumMasters were previously project managers or product managers (although product managers are more likely to transition into the role of product owner). Other ScrumMasters come from a development, testing, or other technical background. As long as an individual has the characteristics that I mentioned earlier and is willing to take on the responsibilities of the role, she can be an effective ScrumMaster.

Some organizations feel that the tech lead or dev lead should be the ScrumMaster. These people in fact might make great ScrumMasters, but they also might *not* be the best choice for the role. People who are in a technical leadership position are there for a reason—they are technically very good at what they do. The ScrumMaster role is not one where that level of technical excellence is exploited to its full potential. Any time technical leaders are doing ScrumMaster work, they are necessarily providing less technical leadership. Making them ScrumMasters, therefore, might adversely affect the technical outcome. I will address later in this chapter whether a development team member can simultaneously fill the ScrumMaster role.

Functional area managers or resource managers can also be successful ScrumMasters if they have the skills to do the job. It would be best if such managers no

longer retained people management responsibility, at least not for members of their Scrum teams. Because a ScrumMaster has no managerial authority, team members might be confused about whether the person is wearing her ScrumMaster or manager hat in a particular instance. I prefer to avoid this situation and not have team members report to the ScrumMaster. In certain organizations, however, it might be unavoidable, so we learn to deal with any potential conflict of interest as best we can.

Is ScrumMaster a Full-Time Job?

Every Scrum team has a ScrumMaster, but is the ScrumMaster a full-time role? Possibly not. A Scrum team that has been working together for an extended period of time and has become highly proficient with Scrum might require less coaching than a new team made up of people who have never worked together and are new to Scrum.

Although the ScrumMaster might need to spend less time with the team day to day as the team matures, the ScrumMaster role remains critical to the success of Scrum within the organization. Usually as the Scrum team's need for a ScrumMaster decreases, the need for the ScrumMaster to focus on broader organizational impediments and to be a change agent throughout the organization value chain increases.

In most cases, the ScrumMaster role remains a significant commitment of time. In those cases where it's not a full-time commitment, some combination of roles may take place.

ScrumMaster Combined with Other Roles

If capacity permits and a single person is both a talented ScrumMaster and development team member, that person may act in both roles. However, this combination could suffer from a conflict of interest when the person tries to wear both hats. For example, what if the person in the combined roles has important ScrumMaster activities (like removing impediments) to perform and also has critical task-level work to do? Because both are important, compromising either will reduce the Scrum team's effectiveness. Complicating the trade-off is the fact that impediments can occur unpredictably and be very time-consuming to address. This makes it even harder to predict how much time a ScrumMaster as team member will actually have available to do task-level work.

There is, however, a different approach that is often better. If a ScrumMaster truly has available capacity, in many cases my preference is to have that person be the ScrumMaster for more than one Scrum team (see Figure 10.4).

Becoming a good ScrumMaster requires acquiring a valuable, not-so-common set of skills. I prefer that a person who has those skills share them with multiple teams rather than spend time performing non-ScrumMaster activities. However, that is just a personal preference. I have seen successful Scrum teams use either of these approaches. There is no generic right or wrong answer, although there might be a right or wrong answer in a specific organizational context.

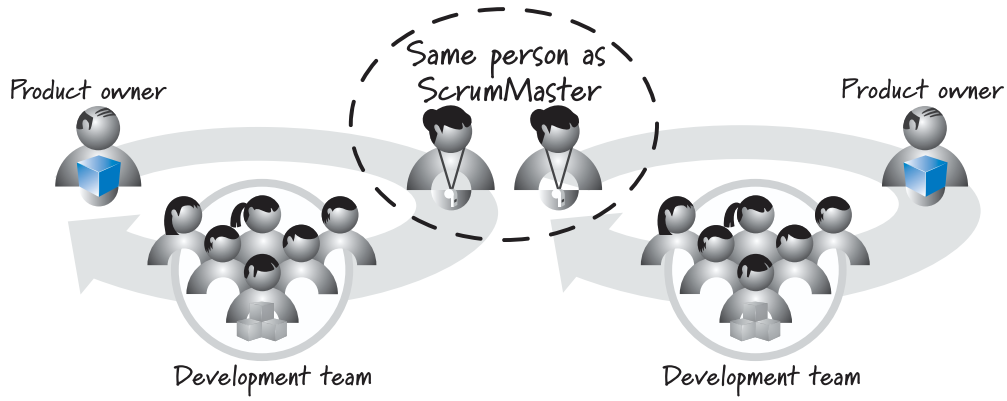


FIGURE 10.4 Same person as ScrumMaster of more than one team

As I mentioned in Chapter 9, one combination of roles that is highly discouraged is having the same person serve as both ScrumMaster and product owner. The ScrumMaster is the coach of the Scrum team, which means the ScrumMaster is the Scrum coach for the product owner. It is hard to be your own coach. In addition, the product owner has real product authority and can make demands on the team. The ScrumMaster frequently acts as the balancing agent between the demands of the product owner and the needs and abilities of the development team. Having the product owner and the ScrumMaster be the same person would add confusion where it could be easily avoided.

Closing

In this chapter I described the ScrumMaster role. I emphasized the ScrumMaster responsibilities as coach, servant leader, process authority, interference shield, impediment remover, and change agent. I then discussed how the ScrumMaster should be knowledgeable about Scrum, ask great questions, patiently wait for the team to solve its problems, collaborate with everyone, protect the team from undue interference, and communicate in a visible and transparent way. Next I described how the ScrumMaster's time is allocated across a sprint to provide a deeper appreciation for this critical role. I concluded by discussing who within the organization should be the ScrumMaster, whether or not the role is full-time, and how the ScrumMaster role might be combined with other Scrum roles.

In the next chapter I will explore the role that the development team plays in Scrum.

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Chapter 11
Development Team

In this chapter I describe the development team role. I begin by discussing five principal responsibilities of this role and conclude by describing ten characteristics that a development team should exhibit.

Overview

Traditional software development approaches define various job types, such as architect, programmer, tester, database administrator, UI designer, and so on. Scrum defines the role of development team, which is simply a cross-functional collection of these types of people. In particular, the development team is one of the three roles on every Scrum team. The development team's members, collectively, have the skills required to deliver the business value requested by the product owner.

The term *development team* may appear to be the wrong label to apply to a team that is composed of more than just developers. Other labels have been used, such as *delivery team*, *design-build-test team*, and just *team*. It's not apparent that any of these labels is more appropriate, less ambiguous, or easier to use than *development team*. For now, the Scrum community has converged on the use of the term *development team*, and I will use that term in this book.

Role-Specific Teams

Many organizations are accustomed to intentionally splitting different job roles into specialized, role-specific teams. These organizations might have one team of designers, one of developers, and another of testers. These teams hand off work to one another when it is complete and more or less function independently of each other.

In Scrum, the development team must do all of the work to produce one or more vertical slices of working product functionality each sprint, including the design, development, integration, and testing of that functionality. Thus, we need a team that is skilled at all of those tasks.

Some organizations try to maintain a separate testing or QA team while doing Scrum. Now, I admit there are times when having a separate team that focuses specifically on testing might be necessary—for example, a regulatory requirement might be that a separate team perform a particular type of testing. However, most of the time there is no such need. Testing should be fully interwoven into the work that takes